

Course: B.Tech

Branch : Information Technology Semester : III

Subject Code & Name: Object Oriented Paradigm with C++ (BTITC304)

Max Marks: 60

Date: 12/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which feature of OOPS described the reusability of code? a. Encapsulation b. Polymorphism c. Inheritance d. Abstraction	L1/CO1	1
2	Which among the following is not a member of the class? a. Data Member b. Const function c. Static function d. Friend Function	L1/CO1	1
3	Which member of the superclass is never accessible to the subclass? a. Public member b. Protected member c. Private member d. All of the mentioned	L1/CO1	1
4	Which of the following statement is NOT true about C++? a. Members of a class are public by default b. A class cannot have the private members c. A structure can have the member functions d. All of the above	L2/CO2	1
5	Select the following which shows the correct constructor syntax. a. ()classname b. classname() c. ~classname() d. None	L2/CO3	1
6	By which of the following can the if-else statement be replaced? a. Bitwise Operator b. Logical Operator c. Conditional Operator d. Arithmetic Operator	L2/CO1	1
7	Which of the following is the correct identifier? a. \$var_name b. VAR_123 c. varname@ d. None	L2/CO2	1
8	The C++ language is _____ object-oriented language. a. Pure Object-oriented b. Not Object-oriented c. Partial Object-oriented d. None of the above	L1/CO1	1
9	Which of the following statements is correct about the class? a. An object is an instance of b. A class is an instance of its c. An object is the same as class d. All are correct	L2/CO2	1

10	Which feature of OOP is exhibited by the function overriding?	d. Inheritance	L2/CO1	1
11	Which of the following refers to the wrapping of data and its functionality into a single individual entity?	a. Modularity b. Abstraction c. Encapsulation d. None	L2/CO1	1
12	How many types of access specifiers are provided in OOP (C++)?	a. 1 b. 2 c. 3 d. 4	L2/CO2	1
Q. 2	Solve the following.			12
A)	Differentiate between procedure-oriented programming and object-oriented programming.		L2/CO1	6
B)	Write a C++ program using a class that asks the user to enter two numbers and calculate the average of them.		L3/CO1	6
Q.3	Solve the following.			12
A)	Explain properties and use of static functions in C++ using a suitable program		L2/CO2	6
B)	Explain the characteristics of the friend function. Write a C++ program to identify odd or even numbers using the friend function.		L3/CO2	6
Q. 4	Solve Any Two of the following.			12
A)	Write a suitable C++ program to explain 'constructor overloading'. Explain the properties of the constructor function.		L3/CO3	6
B)	Using a suitable program, demonstrate the use of 'multiple inheritance' for inheriting 'public' and 'private' members of a parent class in C++.		L3/CO3	6
C)	Explain the concept of abstract class with a suitable C++ program.		L2/CO3	6
Q.5	Solve Any Two of the following.			12
A)	Explain the concept of function overloading. Write a suitable C++ program to calculate the sum of integer numbers given by the user through the function overloading concept.		L3/CO4	6
B)	Explain the concept of operator overloading using member function. Write a C++ program to overload the unary - (subtraction) operator.		L3/CO4	6
C)	Explain the concept of pure virtual function using a suitable C++		L3/CO4	6

	program.		
6	Solve Any Two of the following.		12
A)	Explain the concept of exception handling. Write a suitable C++ program to handle divide by zero exception.	L3/CO5	6
B)	Write a suitable C++ program to demonstrate how MULTIPLE exceptions can be caught in a SINGLE catch block.	L3/CO5	6
C)	Design and write suitable C++ program to re-throw exception (nested throw).	L3/CO5	6
*** End ***			

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